

IO391 Digital I/O Module

Datasheet



Application Information

The IO391 is a mini-PCIe compatible Configurable FPGA I/O module providing a user configurable Xilinx Artix®-7 FPGA with 26 digital I/O channels.

For customer, specific I/O extension or inter-module communication, the IO391 provides 26 FPGA I/O based on the 74LVC2G241 tri-state buffer, allowing user selectable 3.3V and 5V tolerant TTL signaling standards.

The user FPGA is configured by a serial SPI flash. The SPI flash device is in-system programmable. An in-circuit debugging option is available via a JTAG header for read back and real-time debugging of the FPGA design by using the Xilinx Vivado Logic Analyzer.

Technical Information

Mini PCIe Interface	
Mechanical Interface	PCI Express Mini Card Electromechanical Specification, Revision 2.0
Electrical Interface	PCI Express x1 Base Specification, Revision 2.0
On-Board Devices	
User configurable FPGA	IO391: XC7A50T-2 (Xilinx)
SPI-Flash	Macronix 128 Mbit SPI-Flash (contains IO391 Base Firmware) or compatible
User configurable FPGA	IO391: XC7A50T-2 (Xilinx)
SPI-Flash	N25Q128A (Micron) 128 Mbit (contains IO391 FPGA Golden Bitstream)
I/O Interface	
Digital I/O Channels	26 ESD-protected 5 V-tolerant TTL lines
Digital I/O Transceiver	74LVC2G241 (or compatible)

Physical Data	
Power Requirements	+3.3 Vaux: 650 mA typical
Temperature Range	refer to "Thermal Management" chapter
	-40°C to +85°C
MTBF	987.000 h MTBF values shown are based on calculation according to MIL-HDBK-217F and MIL-HDBK-217F Notice 2; Environment: GB 20°C. The MTBF calculation is based on component FIT rates provided by the component suppliers. If FIT rates are not available, MIL-HDBK-217F and MIL-HDBK-217F Notice 2 formulas are used for FIT rate calculation.
Humidity	5 – 95 % non-condensing
Weight	13 g

Table 0-1 : Technical Specification

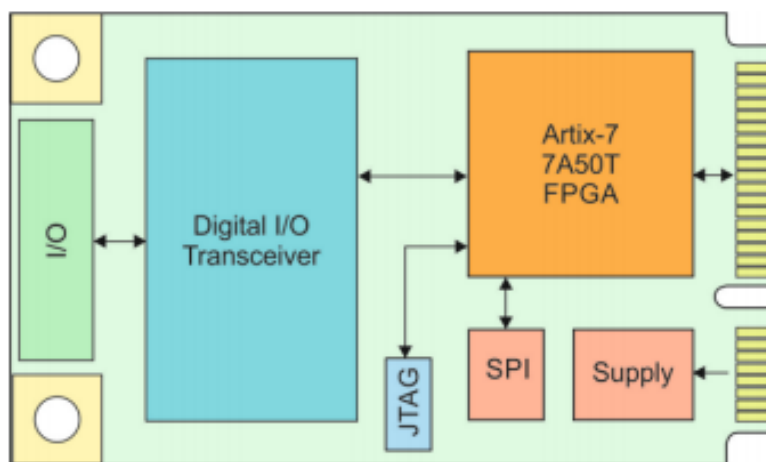


Figure 1 Block diagram